

A Duhamel Formula for Continuously Forced Pantograph Equations

Candidate full-result packet for arXiv:1605.06734

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Status: full solution likely valid; human review required. The formula below removes the source’s analyticity restriction for the stated first-order inhomogeneous pantograph equation and extends the scalar equation to bounded operator coefficients on a Banach space. The proof is complete. Novelty confidence is moderate because an older equivalent Volterra-resolvent formula may exist outside the bounded search.

1 Source target and exact scope

The source is Cheng-shi Liu, *Basic theory of a class of linear functional differential equations with multiplication delay*, arXiv:1605.06734v4 [1]. (The manuscript title printed in the PDF is *Basic theory of a kind of linear pantograph equations*.)

At the transition from pp. 12 to 13, Section 4 says, in part:

“If non-homogenous term is a general smooth function, we do not know how to solve the first order linear pantograph equation...”

It then restricts Theorem 4.1 to analytic forcing in

$$y'(x) = \beta y(\alpha x) + q(x), \quad y(0) = a_0, \quad (1)$$

and derives a Taylor-series formula. The result in this packet answers that specific general-smooth-forcing statement. It does *not* address the paper’s separate Open Problems 3.1 (exact zeros) or 5.1 (a variational principle).

2 Proof intuition

The failed ansatz in the source tries to multiply the homogeneous solution by a variable scalar. The useful replacement is to integrate the equation once. The delay then appears in the scaled Volterra operator

$$(\mathcal{T}f)(x) = \int_0^x f(\alpha s) ds.$$

Its iterates can be calculated exactly. Besides the usual factorial decay of a Volterra operator, the n th iterate gains the factor $\alpha^{n(n-1)/2}$. Thus $I - \mathcal{BT}$ has a globally convergent Neumann inverse on every compact interval, with no smallness assumption on the coefficient. Expanding this inverse against the initial constant and the primitive of the forcing gives the formula below.

4 Power series method for the first order linear non-homogenous pantograph equation

If non-homogenous term is a general smooth function, we do not know how to solve the first order linear pantograph equation since the method of the variation of constant can not be applied in this case. Indeed, for the non-homogenous pantograph equation $y'(x) = \beta y(\alpha x) + q(x)$, the general solution of corresponding homogenous equation is $y(x) = cE_\alpha(\beta x)$. According to the variation of constant method, we can suppose that the special solution of the non-homogenous equation has the form $y(x) = c(x)E_\alpha(\beta x)$, and hence we have $c'(x)E_\alpha(\beta x) + c(x)\beta E_\alpha(\alpha\beta x) = c(\alpha x)\beta E_\alpha(\alpha\beta x) + q(x)$ form which we cannot eliminate two terms $c(x)\beta E_\alpha(\alpha\beta x)$ and $c(\alpha x)\beta E_\alpha(\alpha\beta x)$. Thus, the variation of

12

Figure 1: Source p. 12: the assertion that general smooth forcing is not handled by the scalar variation-of-constants ansatz. The passage continues on p. 13.

constant method is invalid for pantograph equation, and then we must try to find other method. In this section, we try to use power series to solve the first order linear non-homogenous pantograph equation with analytic non-homogenous term. The power series method is simple but so powerful that we use it to find out luckily the structure of the solutions for some important linear pantograph equations.

Theorem 4.1. Consider the following pantograph equation

$$y'(x) = \beta y(\alpha x) + q(x), y(0) = a_0, \quad (43)$$

where β is a constant, and $q(x)$ is a analytic function with expansion

$$q(x) = \sum_{n=0}^{+\infty} q_n x^n.$$

Then its solution is given by

$$y(x) = a_0 E_\alpha(\beta x) + \sum_{n=1}^{+\infty} \left\{ \sum_{k=0}^{n-1} \frac{k! \beta^{n-k-1}}{\alpha^{\frac{k(k+1)}{2}}} q_k \right\} \frac{\alpha^{\frac{n(n-1)}{2}}}{n!} x^n, \quad (44)$$

or another form if the double summations in (44) can exchange each other

Figure 2: Source p. 13: the continuation and Theorem 4.1, which assumes that the forcing is analytic.

3 The full result

For a bounded operator B on a Banach space X , write

$$E_\alpha(Bx) := \sum_{n=0}^{\infty} \alpha^{n(n-1)/2} \frac{x^n B^n}{n!}. \quad (2)$$

This series converges in operator norm for every real (indeed complex) x . For $x < 0$, every integral below is an oriented Bochner integral.

Theorem 1 (Continuous-forcing pantograph resolvent). *Let X be a real or complex Banach space, let $0 < \alpha < 1$, let $B \in \mathcal{L}(X)$, let $q \in C(\mathbb{R}; X)$, and let $a \in X$. There is a unique $y \in C^1(\mathbb{R}; X)$ satisfying*

$$y'(x) = By(\alpha x) + q(x), \quad y(0) = a. \quad (3)$$

It is given by

$$y(x) = E_\alpha(Bx)a + \sum_{n=0}^{\infty} \frac{\alpha^{n(n+1)/2}}{n!} B^n \int_0^x (x-t)^n q(\alpha^n t) dt. \quad (4)$$

The series converges absolutely and uniformly on every compact real interval.

Proof. Fix $R > 0$ and work in $C([-R, R]; X)$. Define

$$(\mathcal{T}f)(x) := \int_0^x f(\alpha s) ds.$$

A direct induction, using oriented integrals when $x < 0$, gives, for $n \geq 1$,

$$(\mathcal{T}^n f)(x) = \frac{\alpha^{n(n-1)/2}}{(n-1)!} \int_0^x (x-t)^{n-1} f(\alpha^n t) dt. \quad (5)$$

For completeness, suppose (5) holds at n . In $\mathcal{T}^{n+1}f$, substitute $u = \alpha t$ in the inner integral and exchange the two integrations. The substitution contributes α , the polynomial kernel contributes α^{n-1} , and the final integration contributes $1/n$. Hence the coefficient is multiplied by α^n/n , exactly changing $\alpha^{n(n-1)/2}/(n-1)!$ into $\alpha^{n(n+1)/2}/n!$. The same computation on $[-R, 0]$, or equivalently the change $x \mapsto -x$, proves the oriented version.

In particular,

$$\|\mathcal{T}^n f\|_{C([-R, R]; X)} \leq \alpha^{n(n-1)/2} \frac{R^n}{n!} \|f\|_{C([-R, R]; X)}. \quad (6)$$

Let $\mathcal{B}$ denote pointwise application of B . The operators $\mathcal{B}$ and $\mathcal{T}$ commute, so (6) implies norm convergence of

$$\sum_{n=0}^{\infty} (\mathcal{B}\mathcal{T})^n$$

on $C([-R, R]; X)$, with no restriction on R or $\|B\|$.

Set

$$Q(x) = \int_0^x q(t) dt, \quad g(x) = a + Q(x).$$

The differential equation (3) is equivalent to the integral equation

$$y = g + \mathcal{B}\mathcal{T}y. \quad (7)$$

Consequently its candidate solution is

$$y = \sum_{n=0}^{\infty} (\mathcal{BT})^n g. \quad (8)$$

The constant part satisfies

$$\mathcal{T}^n a = \alpha^{n(n-1)/2} \frac{x^n}{n!} a. \quad (9)$$

For the forcing part, (5), one more elementary Fubini calculation, and the convention $\mathcal{T}^0 Q = Q$ give, for every $n \geq 0$,

$$(\mathcal{T}^n Q)(x) = \frac{\alpha^{n(n+1)/2}}{n!} \int_0^x (x-t)^n q(\alpha^n t) dt. \quad (10)$$

Indeed, for $n \geq 1$ insert $Q(\alpha^n t) = \alpha^n \int_0^t q(\alpha^n u) du$ into (5); integrating $(x-t)^{n-1}$ from u to x supplies the factor $(x-u)^n/n$. Formula (10) follows, again with the same oriented identity on the negative half-line.

Substitution of (9) and (10) into (8) is exactly (4). If $M_R = \sup_{|s| \leq R} \|q(s)\|$, the norm of its n th forcing term is bounded uniformly for $|x| \leq R$ by

$$M_R R \frac{\alpha^{n(n+1)/2} (\|B\|R)^n}{(n+1)!}. \quad (11)$$

This proves absolute compact-uniform convergence. Equation (8) therefore satisfies (7); since q and y are continuous, the right side of (7) is C^1 and differentiation yields (3).

It remains to prove uniqueness. If z is the difference of two solutions, then $z = \mathcal{BT}z$. Iterating and using (6) gives

$$\|z\|_{C([-R,R];X)} \leq \alpha^{n(n-1)/2} \frac{(\|B\|R)^n}{n!} \|z\|_{C([-R,R];X)} \quad (n \geq 1).$$

The scalar coefficient tends to zero. Thus $z = 0$ on $[-R, R]$. Since R was arbitrary, uniqueness holds on all of $\mathbb{R}$. $\square$

4 Agreement with the source's analytic formula

The following calculation is both a consistency check and the exact link to Theorem 4.1 of the source.

Corollary 1. *In the scalar case $B = \beta$, if $q(x) = \sum_{k \geq 0} q_k x^k$ is analytic, formula (4) has exactly the Taylor coefficients displayed in equation (44) of arXiv:1605.06734v4.*

Proof. The contribution of $q_k t^k$ to the n th forcing term is

$$\beta^n \alpha^{n(n+1)/2 + nk} q_k \frac{k!}{(n+k+1)!} x^{n+k+1},$$

because

$$\int_0^x (x-t)^n t^k dt = \frac{n!k!}{(n+k+1)!} x^{n+k+1}.$$

Put $m = n + k + 1$. The exponent identity

$$\frac{n(n+1)}{2} + nk = \frac{m(m-1)}{2} - \frac{k(k+1)}{2}$$

turns the coefficient of x^m into

$$\frac{\alpha^{m(m-1)/2}}{m!} \sum_{k=0}^{m-1} \frac{k!}{\alpha^{k(k+1)/2}} \beta^{m-k-1} q_k,$$

which is precisely the inhomogeneous coefficient in the source's formula (44). The homogeneous terms also agree by (2). $\square$

5 Adversarial verification

Step check

Step	Verdict	Check
Iterated kernel	Valid	Induction gives the factor $\alpha^{n(n-1)/2}/(n-1)!$; exact monomial checks confirm the indices.
Neumann inversion	Valid	The bound (6) proves operator-norm convergence on every compact interval even when $\ B\ R > 1$.
Forcing kernel	Valid	Fubini yields (10); the $n = 0$ term is exactly Q .
Regularity	Valid	Compact-uniform convergence first gives the integral equation; differentiability then follows from the Banach-valued fundamental theorem of calculus, avoiding unjustified termwise differentiation.
Negative x	Valid	Oriented integrals satisfy the same identities; the equivalent substitution $x \mapsto -x$ changes B, q to $-B, -q(\cdot)$.
Uniqueness	Valid	Iterating the homogeneous integral equation and taking $n \rightarrow \infty$ forces the compact sup norm to vanish.
Match to source	Valid	Corollary 1 recovers equation (44) coefficient by coefficient.

Computer guard checks

The included script `code/verify_formula.py` is not part of the proof. It passed 961 exact exponent identities, 25 exact scalar Taylor coefficients, 38 exact components of a rational 2×2 matrix-coefficient test, and 41 high-precision residual tests on $[-2, 2]$. With 18 retained forcing terms, the maximum residual was $2.3336307 \times 10^{-61}$ and the initial-value error was zero. From the packet directory, the command was:

```
conda run --no-capture-output -n sandbox python code/verify_formula.py
```

6 Novelty check, limitations, and recommendation

Bounded search. The four lightweight indexes for run `fa_banach_001` had no match for the arXiv id or the continuous-forcing formula. External searches used the exact equation and combinations of `pantograph`, `Duhamel`, `variation of constants`, `Volterra resolvent`, and `continuous forcing`. The search also inspected the three works returned by OpenAlex as citing arXiv:1605.06734, the TeX source of arXiv:2303.09543 (which develops general successive approximation but does not display (4)), and metadata/abstract material for classical and recent generalized-pantograph work. No occurrence of the explicit formula (4) was identified. This was a bounded search, not an exhaustive priority claim.

Limitations. The theorem treats the pure proportional-delay equation in the source. It does not include a simultaneous undelayed term $Ay(x)$ with noncommuting operators, neutral terms involving $y'(\alpha x)$, variable dilation, or the paper’s unrelated zero and variational problems. The assumption $q \in C(\mathbb{R}; X)$ can be weakened to local Bochner integrability with an almost-everywhere formulation, but that extension is not claimed here.

Recommendation. The mathematical verdict is **likely valid**, with high proof confidence. Send to human review. The primary mathematical check is the oriented-integral calculation for $x < 0$; the primary bibliographic check is whether an older Volterra or pantograph source already contains the same explicit resolvent.

References

- [1] C.-s. Liu, *Basic theory of a class of linear functional differential equations with multiplication delay*, arXiv:1605.06734v4, 2018 revision; especially Section 4, pp. 12–14.